Spacetime and the Quantum Family Tree

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Abstract

The first two papers in this series established a static correspondence between quantum entanglement and emergent geometry on a Bruhat–Tits tree. *The Emergence of Distance from a Quantum Family Tree* reduced the quenched von Neumann entropy rate for boundary intervals to a single scalar identity — the Lemma D conjecture

$$S_{VN} = \left(c_{VN}/3\right) \cdot k, \qquad c_{VN} = m^2 \cdot c_{RT}, \qquad m^2 = \left(1 - \eta_2\right)/2 = 3/10$$

for the binary Haar tree, supported numerically at 10.34σ and independently confirmed by a 156-qubit IBM Heron quantum run. *The Emergence of Geometry from a Quantum Family Tree* extended this to p-ary trees with d -dimensional bonds, establishing exact Weingarten formulas and proving the structural Reduction Theorem, with the universality conjecture

$$w_2(p, d) = \left(1 - \eta_2\right)/2$$

together with four unconditional structural theorems: fixed-point convergence, the first law at all scales, the geometric precondition, and the boundary-mode decomposition. The 1/2 factor in w_2 is derived unconditionally from the symmetry of the two interval endpoints; the remaining scalar identity $\chi \cdot V^* = (1/4)\log_2(1/\eta_2)$ is conjectural at all (p,d).

Those results describe what the geometry looks like. This paper asks what happens when the tree lives.

We show that the Quantum Family Tree and its emergent spacetime admit a precise structural identification—they are not two objects connected by a dictionary but one object described in two languages. We establish this through

six structural theorems: (i) tree ancestry defines an exact causal structure with exponential perturbation screening at rate η_2; (ii) tree growth is cosmological expansion, with de Sitter scale factor

$$a(t) = p^t, \qquad H = \ln p;$$

(iii) gate perturbations produce entropy geometry deformations governed by a retarded screened response equation

$$\delta s_A = \sum_{n \in Past(A)} G(A, n)\, \delta j_n$$

with massive propagator $m_{eff} = \frac{1}{2} \log\left(1/\eta_2\right)$; (iv) the holographic RG flow has the Haar measure as an exponentially stable attractor with a monotone c-function; (v) the modular Hamiltonian is computed explicitly at $k = 1$; (vi) these are assembled into the Conjoined Theorem (Theorem 10), a formal duality between circuit dynamics and spacetime structure. The correspondence is linearized and structural; the full Jacobson-style dynamical equation and the global reconstruction conjecture are formulated as open problems.

Introduction

This is the third paper in the Quantum Family Tree (QFT) series. The first two established a precise quantitative link between random quantum circuits on a Bruhat–Tits tree and emergent geometry:

The Emergence of Distance from a Quantum Family Tree established that Haar-random $U(4)$ gates on a binary tree produce a boundary state whose entanglement entropy grows linearly with the geodesic depth k of the interval, with coefficient $c_{VN} = m^2 \cdot c_{RT}$, where $m^2 = \left(1 - \eta_2\right)/2 = 3/10$ (conjectured; Lemma D) and $\eta_2 = 2/5$ is the (proved) Haar-averaged purity eigenvalue. The structural reduction—Weingarten eigenvalue, mode budget, annihilation lemma, exchange symmetry, cut averaging—reduces the entropy-rate theorem to a single scalar identity (Lemma D). Lemma D remains analytically open; it is supported numerically at 10.34σ by classical Monte Carlo (N=2000, D=20–150) and independently confirmed by a 156-qubit IBM Heron quantum run.

The Emergence of Geometry from a Quantum Family Tree generalized this to p-ary trees with d-dimensional bonds, proving exact Weingarten purity formulas for all (p, d), the fixed-point theorem $\rho^* = I/d$ with L^2 contraction at rate η_2, the first law $\delta\langle H_A\rangle = \delta S$ at all scales, the geometric precondition with exponentially small finite-depth corrections, and the boundary-mode decomposition $\Delta_k = J_L + J_R + \varepsilon_k$. The 1/2 factor in the universality conjecture $w_2 = (1-\eta_2)/2$ is derived unconditionally from the two-endpoint symmetry; the full identity remains conjectural.

Both papers treated the tree as a static object: a fixed depth D, a fixed set of gates, a frozen boundary state. The emergent geometry was a photograph—distance at one moment.

This paper makes the tree breathe.

The central claim is stronger than emergence. Papers 1 and 2 showed that spacetime *comes out of* the tree. This paper shows that spacetime and the tree are *the same object*. Perturb the tree—change a gate, add a generation—and the geometry deforms. The deformation respects the causal structure defined by tree ancestry, propagates with exponential screening set by the same eigenvalue η_2 that controls the static geometry, and satisfies a retarded response equation with a massive propagator whose screening mass is determined by the transfer gap. The tree does not model spacetime. The tree is spacetime. They are conjoined.

Outline.

Section 2 defines the dynamical QFT model. Section 3 proves the causal structure theorems. Section 4 establishes time as tree growth, including de Sitter expansion. Section 5 proves the back-reaction theorems. Section 6 derives the linear response equation. Section 7 computes the modular Hamiltonian at $k = 1$. Section 8 develops the holographic RG structure. Section 9 states and proves the Conjoined Theorem. Section 10 is the discussion. Section 11 is the conclusion.

The Dynamical QFT Model

Static model (review)

Let T_D denote the complete p-ary rooted tree of depth D, with p^D boundary (leaf) sites and $\left(p^D - 1\right)/(p - 1)$ internal nodes. Each internal node n carries an independent Haar-random unitary $V_n \in U\left(d^p\right)$. At each node, the gate V_n acts on $\left(\rho_{carry} \otimes |0\rangle\langle 0|^{\otimes(p-1)}\right)$ and the p output qudits are routed to the p children. The boundary state ρ_∂ is obtained by tracing out all internal degrees of freedom.

For a boundary interval A of p^k consecutive leaves, the reduced density matrix ρ_A is computed via the lazy-tree Choi-matrix recursion . The von Neumann entropy satisfies $S_{VN}(A) \approx \left(c_{VN}/3\right) \cdot\ k$ in the deep-tree limit.

Dynamical extension

The **dynamical QFT** is defined by a growth process. At each discrete time step $t\ =\ D$, the tree T_D grows by one generation: each leaf of T_D sprouts p children via fresh independent Haar-random gates, producing T_{D+1} with p^{D+1} boundary sites.

The tree grows at the **bottom**. The root is unchanged. All existing gates are unchanged. The only new randomness is at depth D. This structural point—that growth preserves the entire existing architecture—is the foundation of the dynamical theory.

Causal Structure from Tree Ancestry

Definitions

For a boundary interval A, define:

- $Path(A)$ = set of nodes on the root-to-$LCA(A)$ path (inclusive)
- $Sub(A)$ = set of nodes in the subtree rooted at $LCA(A)$
- $Past(A) =\ Path(A) \cup Sub(A)$

Causal factorization

Theorem 1 (Causal factorization). *The reduced density matrix ρ_A depends only on $\{V_n : n \in Past(A)\}$ and ρ_{root}. For any node $m \notin Past(A)$ and any replacement gate $V_m' \in U\left(d^p\right)$: $\rho_A\left[\{V_n\}_{n\neq m} \cup \{V_m'\}\right] = \rho_A\left[\{V_n\}\right]$.*

Proof. The Choi recursion computes ρ_A by composing channels from root to boundary. At each node n on $Path(A)$, the channel Φ_{V_n} acts on $\left(\rho_n \otimes |0\rangle\langle 0|\right)$ and produces p children. The child heading toward $LCA(A)$ receives the partial trace over the other $p - 1$ children. Those $p - 1$ sibling subtrees are fully traced out at node n—no gate below them enters the computation of ρ_A. Within $Sub(A)$, all gates participate. Outside $Past(A)$, every gate either lives in a sibling subtree fully traced out at some node along $Path(A)$, or lives in a subtree disconnected from A entirely. In both cases, the partial trace eliminates any dependence. □

Perturbation screening

Theorem 2 (Perturbation screening). *For a gate perturbation $V_n \to V_n + \delta V$ at node n:*

1. *If $n \notin Past(A)$, then $\delta S_{VN}(A) = 0$ exactly.*
2. *If $n \in Path(A)$ at depth d_n, with $LCA(A)$ at depth d_A, then $\left|\delta S_{VN}(A)\right| \le C(k) \cdot \eta_2^{\left(d_A - d_n\right)/2}$.*
3. *If $n \in Sub(A)$, then $\left|\delta S_{VN}(A)\right| = O(1)$.*

Proof. Part (a) is immediate from Theorem 1. For part (b), the perturbation δV at node n produces a carry perturbation that propagates through $\left(d_A - d_n\right)$ independent Haar-random gates. By the fixed-point theorem , $E\left[\| T_{d_A} \cdots T_{d_n+1} \delta r\|^2\right] = \eta_2^{d_A - d_n} \| \delta r\|^2$. The boundary entropy is Lipschitz in the carry with a k-dependent constant $C(k)$ from the boundary-mode decomposition . Part (c): the gate directly participates in producing ρ_A. □

Light cones

A perturbation at node n (depth d) has a forward cone containing all p^{D-d} descendants—expanding as $p^{\Delta t}$, exponentially, not linearly. This is p-adic causal structure. There are no shortcuts: if $n \notin Past(A)$, the effect on ρ_A is exactly zero.

Time as Tree Growth

Growth theorem

Theorem 3 (Growth). *For a fixed-LCA interval with $k = D - d$ levels:*

$$E\left[S_{VN}^{(D+1)}\left(A^{(D+1)}(n)\right)\right] = E\left[S_{VN}^{(D)}\left(A^{(D)}(n)\right)\right] + h_{VN} + O\left(\left(\frac{\eta_4}{\eta_2}\right)^k\right),$$

where $h_{VN} = c_{VN}/3$ is the asymptotic entropy per level and $\eta_4/\eta_2 = 13/28$ is the subleading correction rate.

Proof. Since all gates above depth D are unchanged, the carry at every existing node is identical in T_{D+1} and T_D. The boundary state of $A^{(D+1)}$ is obtained from that of $A^{(D)}$ by applying one more layer of Haar-random channels at the old leaves. The entropy increment $\Delta_k = S(k + 1) - S(k)$ approaches h_{VN} as $k\to\infty$, with corrections decaying at rate $\eta_4/\eta_2 = 13/28$ per level . □

Co-moving stability

Theorem 4 (Co-moving stability). *Fix interval size p^k. Then:*

$$\left|E\left[S_{VN}^{(D+1)}(co - moving, k)\right] - E\left[S_{VN}^{(D)}(k)\right]\right| = O\left(\eta_2^{D-k}\right).$$

Proof. A co-moving interval uses k levels in both trees. The carry entering the LCA has contracted through $D - k$ (resp. $D + 1 - k$) steps in T_D (resp. T_{D+1}), both within $O\left(\eta_2^{D-k}\right)$ of stationary by the fixed-point theorem. □

De Sitter expansion

Define cosmological time $t = D$ and scale factor $a(t) = p^t$. Then $a(t + 1) = p \cdot a(t)$, giving $\dot{a}/a = ln p$. The expansion is exponential with Hubble rate $H = ln p$. For the binary tree, $H = ln 2 \approx 0.693$.

The arrow of time and the second law

The tree has a natural direction: root → leaves. Time is genealogical depth. The arrow of time is the arrow of descent. For any fixed-LCA interval, S_{VN} grows monotonically with tree depth (Theorem 3, since $h_{VN} > 0$). The second law is a direct consequence: each new generation injects fresh correlations through the carry channel.

Back-Reaction: Gate Perturbations Deform Geometry

Single-gate perturbation

This is Theorem 2. A single gate perturbation produces an entropy perturbation that is exactly zero outside $Past(A)$, exponentially screened on $Path(A)$, and $O(1)$ on $Sub(A)$.

Collective slice perturbation

Theorem 5 (Collective slice perturbation). *Let A be a boundary interval with LCA at depth* d_A*. Consider a uniform perturbation of all gates at depth* d*:* $V_n \to V_n + \delta V_n$ *for* $n \in L_d$*, with* $\left|\delta V_n\right| \leq \varepsilon$*. Then:*

1. $d < d_A$ *(ancestor slice):* $\left|\delta S_{VN}(A)\right| \leq C(k) \cdot \varepsilon \cdot \eta_2^{\left(d_A - d\right)/2}$.
2. $d = d_A$ *(LCA slice):* $\left|\delta S_{VN}(A)\right| = O(\varepsilon)$.
3. $d > d_A$ *(descendant slice):* $\left|\delta S_{VN}(A)\right| = O(\varepsilon)$ *for fixed interval size.*

For translation-invariant perturbations, co-moving intervals experience the same expected entropy shift up to $O\left(\eta_2^{D-k}\right)$ *corrections.*

Proof. Case I: the depth-d slice intersects $Past(A)$ in exactly one node on $Path(A)$; all others have zero effect by Theorem 1. Screening follows from Theorem 2(b). Case II: same argument with the unique relevant node equal to $LCA(A)$. Case III: since ρ_A lives in a fixed finite-dimensional Hilbert space, the Fréchet derivative of the entropy is bounded, and the total variation is $O(\varepsilon)$ independent of D. Translation invariance plus co-moving stability (Theorem 4) gives position-independence. □

The Linear Response Equation

Response kernel

Theorem 6 (Linear response). *For infinitesimal gate perturbations* $V_n \to V_n + \varepsilon\dot{V}_n$, *define*

$$\delta s_A := (d/d\varepsilon)\, S_{VN}(A)|_{\varepsilon=0}. \text{ Then: } \delta s_A = \sum_{n\in Past(A)} G(A,n)\,\delta j_n$$

where δj_n *denotes the linearized source induced by* $\dot{V}_n$ *on the outgoing carry channel, and the Green's function satisfies:*

1. ***Causality:*** $G(A,n) = 0$ *if* $n \notin Past(A)$.
2. ***Path screening:*** $|G(A,n)| \le C(k)\cdot \eta_2^{(d_A - d_n)/2}$ *for* $n \in Path(A)$.
3. ***Subtree locality:*** $|G(A,n)| \le C_A$ *for* $n \in Sub(A)$.
4. ***Co-moving homogeneity:*** *For translation-invariant perturbations,* $E[\delta s_A]$ *depends only on k, up to* $O\left(\eta_2^{D-k}\right)$.

Proof. Property (1) is the derivative of Theorem 1. Property (2) is the derivative of Theorem 2(b). Property (3) follows from Theorem 5 (Case III). Property (4) combines Theorem 4 with translation invariance. □

Depth-slice form

Define the depth-d source $J_d(A) := \sum_{n\in L_d \cap Past(A)} \delta j_n$. Then:

$$\delta s_A = \sum_{d=0}^{D} K(A;d)\, J_d(A),$$

with $K(A;d) = 0$ outside $Past(A)$, $K(A;d) \sim \eta_2^{(d_A-d)/2}$ for $d < d_A$ (Yukawa-screened ancestor slices), and $K(A;d) = O(1)$ for $d \geq d_A$ (direct descendant sources).

Massive propagator

The path kernel decays as $G(A,n) \sim exp\left(-m_{eff} \cdot \left(d_A - d_n\right)\right)$ with screening mass

$$m_{eff} = \frac{1}{2} log\left(1/\eta_2\right) \approx 0.458.$$

The same spectral gap that determines the conjectured holographic mass-squared (conditional on Lemma D) $m^2 = \left(1 - \eta_2\right)/2 = 3/10$ controls the screening of the response kernel.

The Modular Hamiltonian at $k = 1$

For a $k = 1$ interval A (two boundary sites), the modular Hamiltonian is $H_A = - ln\rho_A$, acting on the support of ρ_A.

The following statistics are empirical estimates over Haar realizations (500 trees, $D = 30$), serving to characterize the typical modular spectrum.

Theorem 7 (Aligned modular Hamiltonian). *For an aligned $k = 1$ interval, $rank\left(\rho_A\right) = 2$. The modular Hamiltonian acts on the 2D support: $H_A = U \cdot diag\left(-ln\lambda_1, -ln\lambda_2\right) \cdot U^\dagger$.*

Measured: $\lambda_1 = 0.782 \pm 0.092$, $\lambda_2 = 0.218 \pm 0.092$. $S_{VN} = 0.719 \pm 0.177$ bits. Purity 0.676 (target $p_{stat} = 2/3$). Modular gap $\Delta h = 1.38 \pm 0.62$ nats.

Theorem 8 (Straddling modular Hamiltonian). *For a straddling $k = 1$ interval, $rank\left(\rho_A\right) = 4$. The modular Hamiltonian is a full 4×4 Hermitian matrix.*

Measured: $\mu_1 = 0.612 \pm 0.099$, $\mu_2 = 0.239 \pm 0.062$, $\mu_3 = 0.108 \pm 0.045$, $\mu_4 = 0.041 \pm 0.027$. Spectral width $h_4 - h_1 = 2.97 \pm 0.98$ nats. The entropy increment from straddling is $\Delta = 0.693$ bits—the microscopic unit of entropy production.

Holographic Renormalization Group

RG dictionary

The tree depth maps to energy scale: root ($d = 0$) is the IR; leaves ($d = D$) are the UV. Each gate layer is one RG step. The carry state ρ_d is the running coupling at scale d.

Haar attractor and c-theorem

Theorem 9 (RG attractor and c-theorem). *The Haar measure is an exponentially stable RG attractor with convergence rate* η_2. *The* L^2 *carry information* $c(d) = E\left[\| r_d\|^2\right]$ *satisfies* $c(d + 1) = \eta_2 c(d)$ *exactly, and is strictly monotonically decreasing to zero.*

Proof. By the fixed-point theorem , $E\left[\| \rho_d - I/d\|^2\right] = \eta_2^d \cdot \|\rho_0 - I/d\|^2$. The c-function identity follows from $E\left[T^\dagger T\right] = \eta_2 I$. □

Two sources of entropy

In the deep-tree limit, entropy production has two sources. *Source 1 (transient):* the carry's initial-condition information drains into boundary correlations, decaying as η_2^d. *Source 2 (stationary):* at the fixed point $\rho^* = I/d$, each gate mixes a maximally-mixed carry with a fresh $|0\rangle$ ancilla by a random unitary, creating entanglement between children *ex nihilo*. In the asymptotic regime, Source 2 is the sole contributor to h_{VN}.

Scale invariance and MERA connection

The entropy increment $\Delta_k \to h_{VN}$ is independent of k, with corrections $O\left((13/28)^k\right)$. The effective central charge $c_{eff} = 3h_{VN} = c_{VN}$ is constant. The QFT model is architecturally equivalent to a MERA tensor network with Haar-random gates replacing variational ones. The randomness produces the Jensen gap $\gamma = 21/20$ and the suppression $m^2 = 3/10$ relative to the RT bound (conditional on Lemma D of Paper 1).

The Conjoined Theorem

Theorem 10 (Conjoined Tree–Spacetime Correspondence). *Consider the Quantum Family Tree defined by a rooted p-ary tree with i.i.d. Haar-random gates at each node. In the stationary (deep-tree) regime:*

1. ***Geometry from entanglement.*** $S_{VN}(A) = \left(c_{VN}/3\right) \cdot k + O\left(\left(\eta_4/\eta_2\right)^k\right)$, *with* $c_{VN} = m^2 c_{RT}$ *and* $m^2 = \left(1 - \eta_2\right)/2$. *[Papers 1, 2; conditional on Lemma D, numerically supported at 10.34σ and confirmed by 156-qubit IBM Heron quantum run]*

2. ***Causal structure from ancestry.*** $Past(A)$ *is the exact causal past. Perturbations outside* $Past(A)$ *have zero effect. Perturbations on* $Path(A)$ *are screened at rate* $\eta_2^{\left(d_A - d_n\right)/2}$. *[Theorems 1, 2]*

3. ***Dynamical response law.*** $\delta s_A = \sum_{n \in Past(A)} G(A, n)\, \delta j_n$, *with causality, path screening, and subtree locality. [Theorem 6]*

4. ***Time as tree growth.*** *Entropy grows as* h_{VN} *per generation. Co-moving intervals are stable. The scale factor* $a(t) = p^t$ *gives de Sitter expansion. [Theorems 3, 4]*

5. ***RG structure.*** *The carry* c*-function* $c(d) = \eta_2^d$ *is monotonically decreasing. The Haar measure is an exponentially stable attractor. [Theorem 9]*

6. ***Screening as horizon.*** *Perturbations decay with mass* $m_{eff} = \frac{1}{2} log\left(1/\eta_2\right)$, *defining screening length* $\ell_{screen} = 1/log\left(1/\eta_2\right)$. *[Theorems 2, 6]*

Proof. Each component follows from earlier theorems: (i) from the entropy scaling and reduction theorem (Papers 1–2, conditional on Lemma D); (ii) from Theorems 1–2 [unconditional]; (iii) from Theorem 6 [unconditional]; (iv) from Theorems 3–4 [unconditional]; (v) from Theorem 9 [unconditional]; (vi) from the exponential decay of the response kernel [unconditional]. Items (ii)–(vi) hold outright; item (i) inherits the Lemma D conditional status. Their conjunction defines a consistent structure satisfying all listed properties, with the quantitative coefficient in (i) subject to closure of Lemma D. □

Scope and limitations

Theorem 10 is a linearized and structural correspondence. Four extensions remain open: (0) Lemma D itself — the sector-weight identity $w_2 = (1-\eta_2)/2$, on which the quantitative coefficient in item (i) depends; (1) a full dynamical equation for the source field δj_n analogous to the Einstein equation (the Jacobson inversion program); (2) the fluctuation-dissipation relation $\gamma(q)$ for $q \geq 3$; (3) global invertibility of the map $\{V_n\}/gauge \to \{S(A)\}_A$ (the reconstruction conjecture). Items (1)–(3) do not affect the validity of Theorem 10. Item (0) affects the quantitative value of the entropy coefficient in item (i) but not the structural claim that geometry emerges from entanglement at the conjectured rate.

Discussion

Conjoining vs. emergence

Papers 1 and 2 used the language of emergence: spacetime *comes out of* the tree. This paper argues for a stronger identification. The tree and its spacetime are not cause and effect but two descriptions of a single mathematical object, connected by exact dualities at every level—causal, dynamical, thermodynamic, and renormalization-theoretic. The word "conjoining" captures this: neither is prior; they are born together.

The screening mass

The effective screening mass $m_{eff} = \frac{1}{2} log\left(1/\eta_2\right)$ appears in the linear response kernel and determines the horizon scale. This is not the naive graph-Laplacian eigenvalue (falsified at $p \geq 3$) but a quantity determined by the Haar-averaged purity contraction—the same spectral gap that controls the conjectured static mass (conditional on Lemma D) $m^2 = \left(1 - \eta_2\right)/2$.

Vacuum entanglement creation

The most striking result of the RG analysis is that the asymptotic entropy rate h_{VN} is not a transfer of pre-existing information but a creation of new correlations from the quantum vacuum—an information-theoretic analog of Hawking radiation.

Conclusion

We have established that the Quantum Family Tree and its emergent spacetime are conjoined at the structural level: tree ancestry is causal structure, tree growth is cosmological expansion, carry contraction is the horizon, and the entropy response to gate perturbations satisfies a retarded screened dynamical law. These results are collected in the Conjoined Theorem (Theorem 10). The structural claims (items 2–6) are proved unconditionally; the quantitative entropy coefficient (item 1) is conditional on Lemma D of Paper 1, which is numerically supported at 10.34σ classical Monte Carlo and confirmed independently by a 156-qubit IBM Heron quantum run.Four problems remain open: Lemma D itself (the sector-weight identity), the full Jacobson-style dynamical equation, the fluctuation-dissipation relation at higher replica order, and the global reconstruction conjecture. Lemma D's closure would upgrade the entropy coefficient in Theorem 10(1) from conjectured to proved; the remaining three problems concern extensions of the framework and do not affect the structural correspondence.The tree is not a model of spacetime. The tree is spacetime—a pulsing, living, ever-expanding entity whose growth is time and whose entanglement is distance. They are conjoined.

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